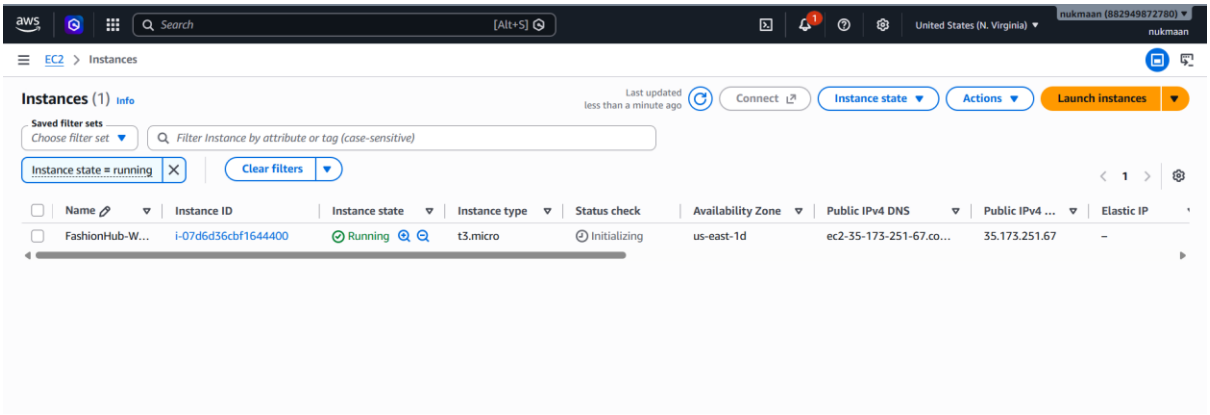
STEP 1 – Launch EC2 Instance

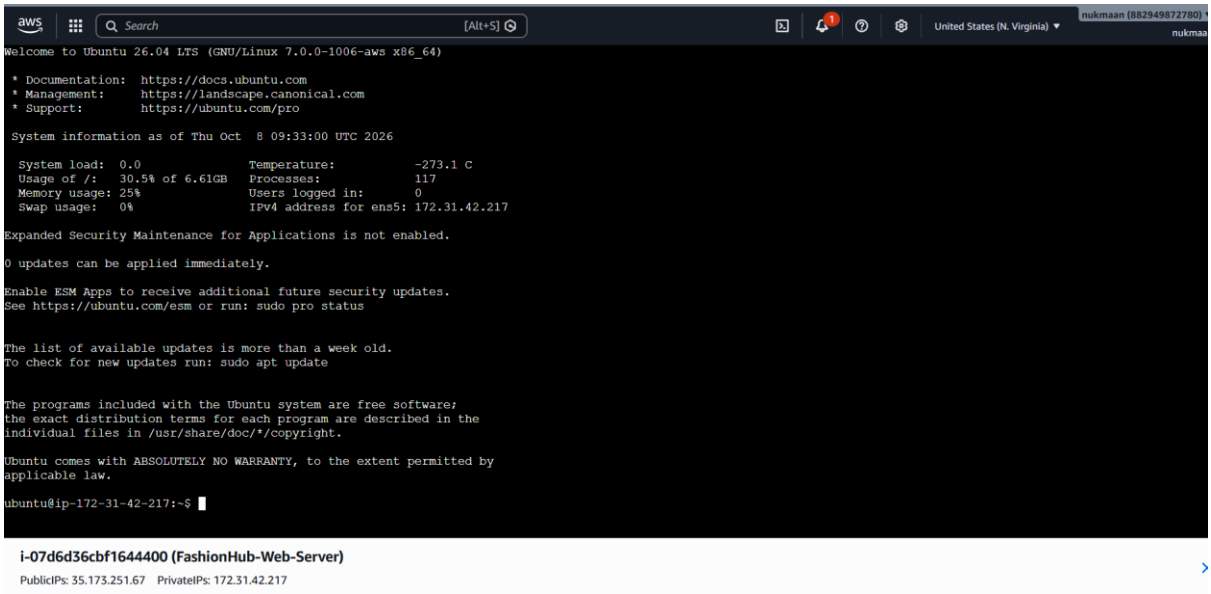
1. Open AWS Console.
2. Select us-east-1.
3. Open EC2.
4. Click **Launch instance**.
5. Select the required AMI and instance type.
6. Configure the required settings.
7. Launch the instance.

Screenshot:



Step 2 – Connect to the EC2 Instance

1. Navigate to AWS Console → EC2 → Instances.
2. Select the FashionHub-Web-Server instance.
3. Click Connect.
4. Select EC2 Instance Connect.
5. Enter the username as ubuntu.
6. Click Connect.
7. A terminal window opens, confirming a successful connection to the Ubuntu EC2 instance.



Step 3 – Installation of Apache Web Server

1. Update the Ubuntu package repository.
2. Install the Apache2 web server.
3. Verify that the Apache service is running.

Commands:

sudo apt update

sudo apt install apache2 -y

sudo systemctl status apache2

Expected Output:
active (running)

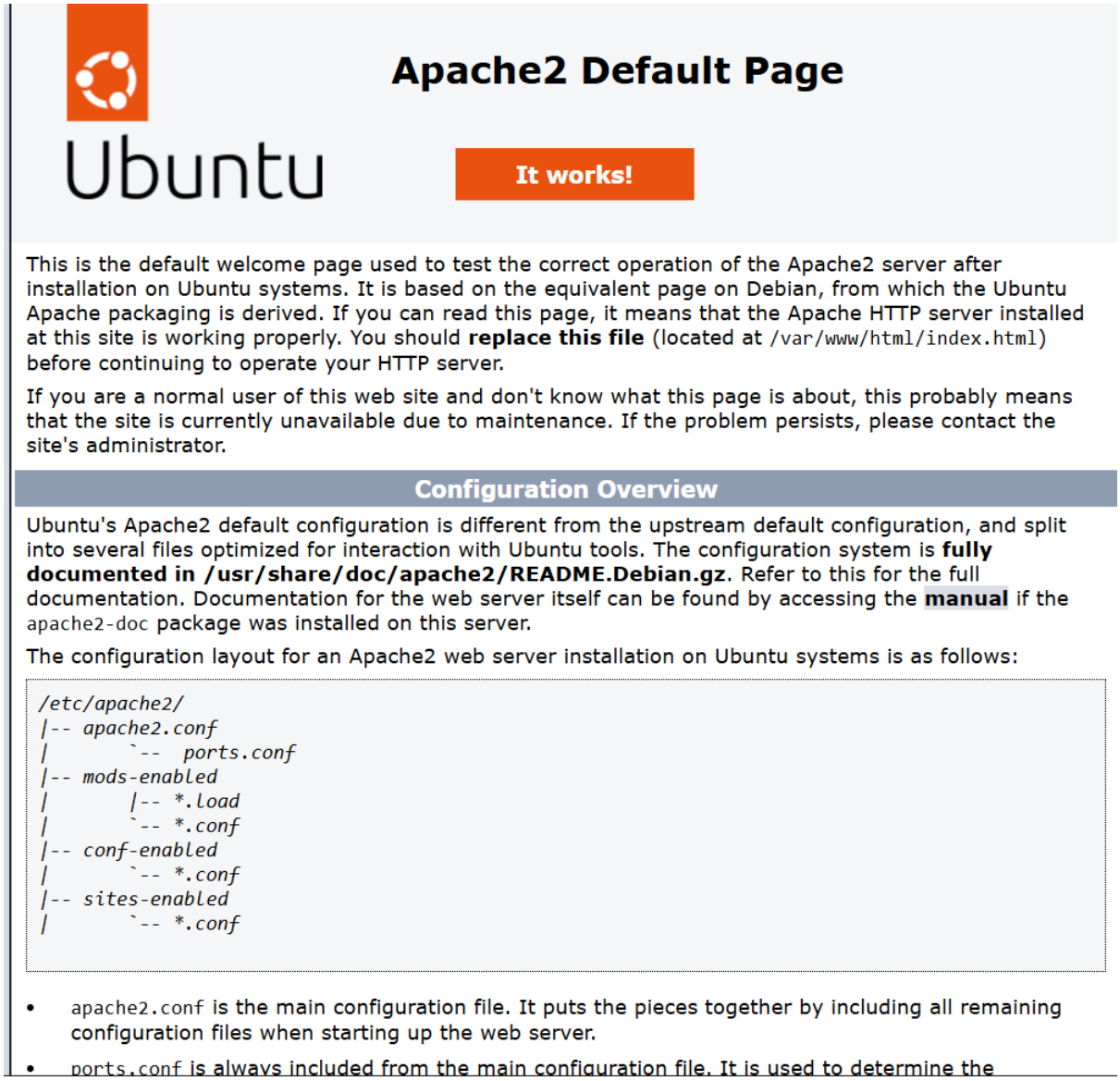
```
No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-42-217:~$ sudo systemctl status apache2
● apache2.service - The Apache HTTP Server
     Loaded: loaded (/usr/lib/systemd/system/apache2.service; enabled; preset: ena>
     Active: active (running) since Thu 2026-10-08 09:40:43 UTC; 25s ago
 Invocation: 0c843b33604543aebb0ffdeff28ce40d
       Docs: https://httpd.apache.org/docs/2.4/
   Main PID: 2396 (apache2)
     Status: "Total requests: 0; Idle/Busy workers 100/0;Requests/sec: 0; Bytes se>
      Tasks: 55 (limit: 627)
     Memory: 5.8M (peak: 5.8M)
        CPU: 96ms
     CGroup: /system.slice/apache2.service
             ├─2396 /usr/sbin/apache2 -k start -DFOREGROUND
             ├─2412 /usr/sbin/apache2 -k start -DFOREGROUND
             └─2413 /usr/sbin/apache2 -k start -DFOREGROUND

Oct 08 09:40:43 ip-172-31-42-217 systemd[1]: Starting apache2.service - The Apache>
Oct 08 09:40:43 ip-172-31-42-217 systemd[1]: Started apache2.service - The Apache >
lines 1-17/17 (END)
```

i-07d6d36cbf1644400 (FashionHub-Web-Server)

PublicIPs: 35.173.251.67 PrivateIPs: 172.31.42.217



Step 4 – Create an S3 Bucket

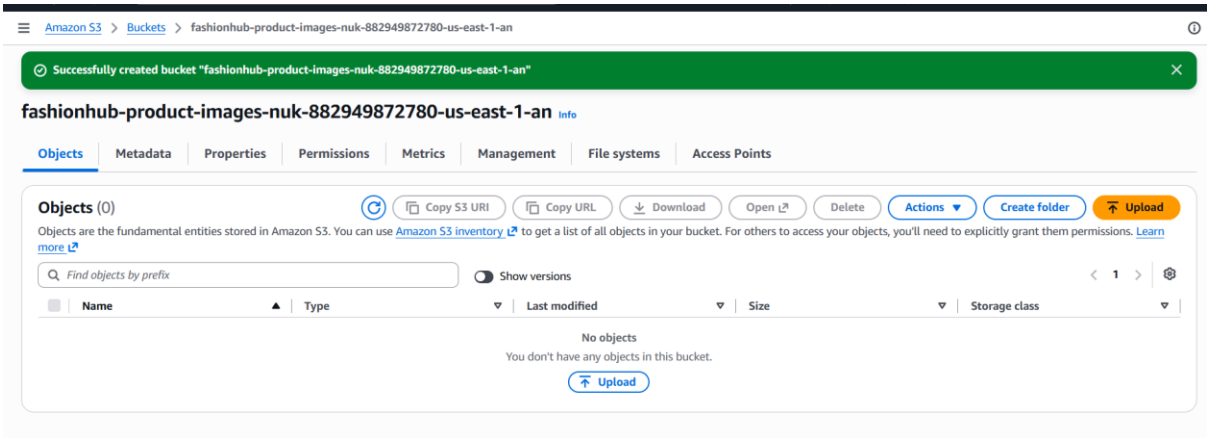
1. Open Amazon S3 from the AWS Management Console.
2. Click Create bucket.
3. Select US East (N. Virginia) – us-east-1.
4. Enter a globally unique bucket name, for example:

fashionhub-product-images-yourname

5. Select Bucket owner enforced under Object Ownership.
6. Keep Block all public access enabled.
7. Enable Bucket Versioning.

8. Select SSE-S3 for default encryption.
9. Keep the remaining settings as default.
10. Click Create bucket.

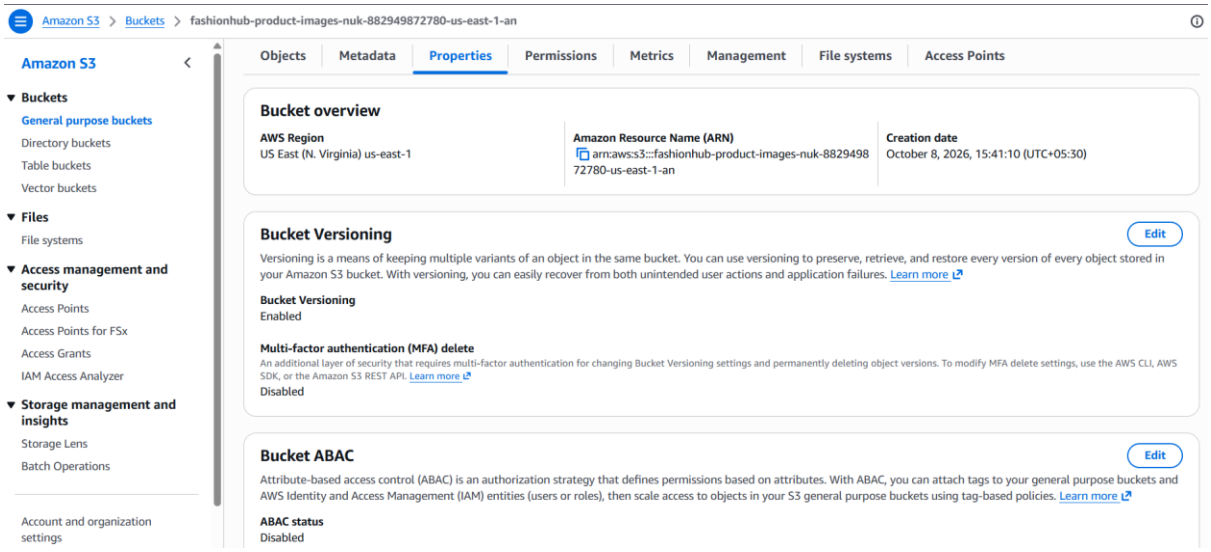
SCREENSHOT:



Step 5 – Enable and Verify S3 Versioning

1. Open the FashionHub S3 bucket.
2. Navigate to the Properties tab.
3. Verify that Bucket Versioning is enabled.
4. Versioning allows previous versions of objects to be retained when an object is overwritten or deleted**.**

SCREENSHOT :



Step 6-Configure S3 Block Public Access:

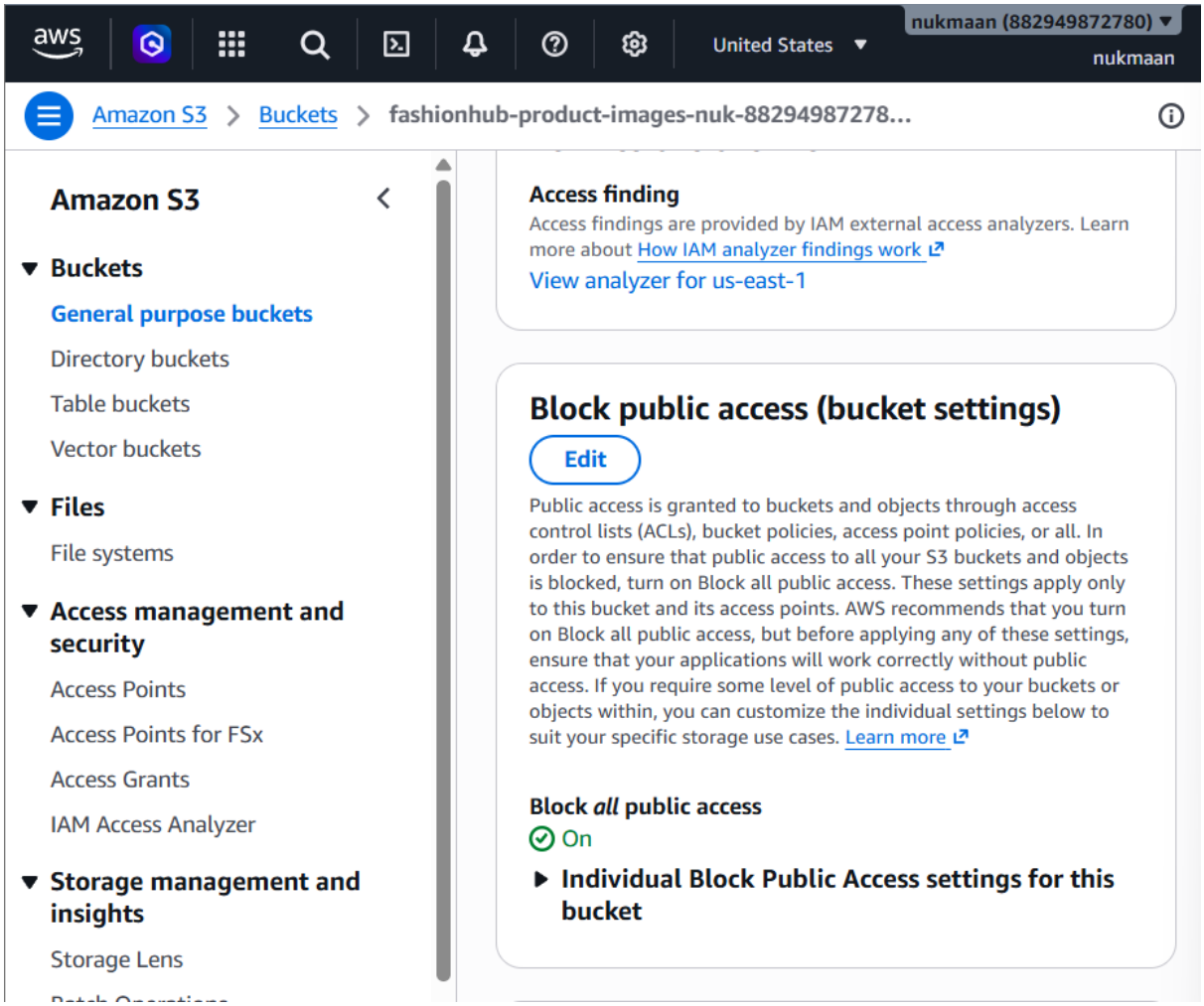
1.Open the FashionHub S3 bucket.

2.Navigate to the Permissions tab.

3.Locate Block public access settings.

4.Verify that Block all public access is enabled.

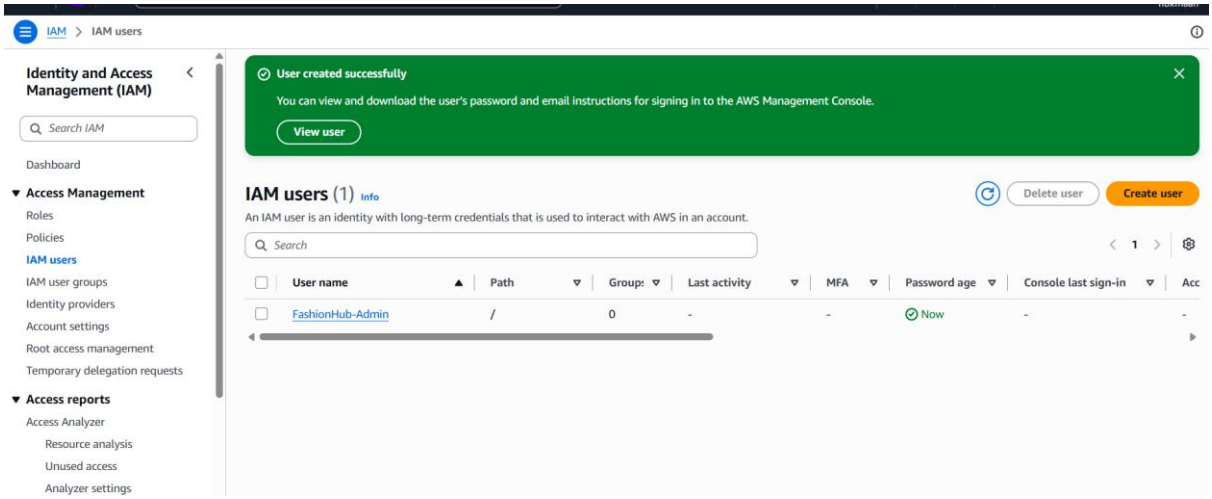
SCREENSHOT:



Step 7– Configure IAM Access Control

1. Open the IAM service.
2. Create an IAM user named FashionHub-Admin.
3. Configure console access if required.
4. Attach the required S3 permission policy.
5. Create the IAM user.

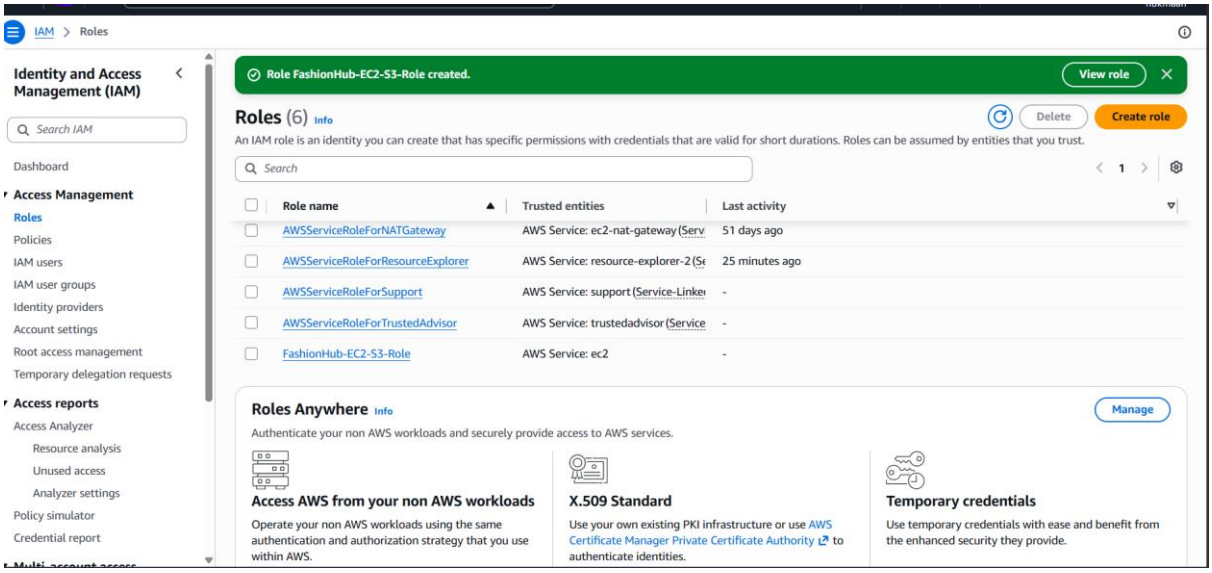
SCREENSHOT:



Step 8 – Assign IAM Role to EC2

An IAM role was created for the EC2 instance with read-only access to Amazon S3. The role was then attached to the FashionHub EC2 instance. This allows the web application to access S3 without storing AWS access keys on the server.

SCREENSHOT:



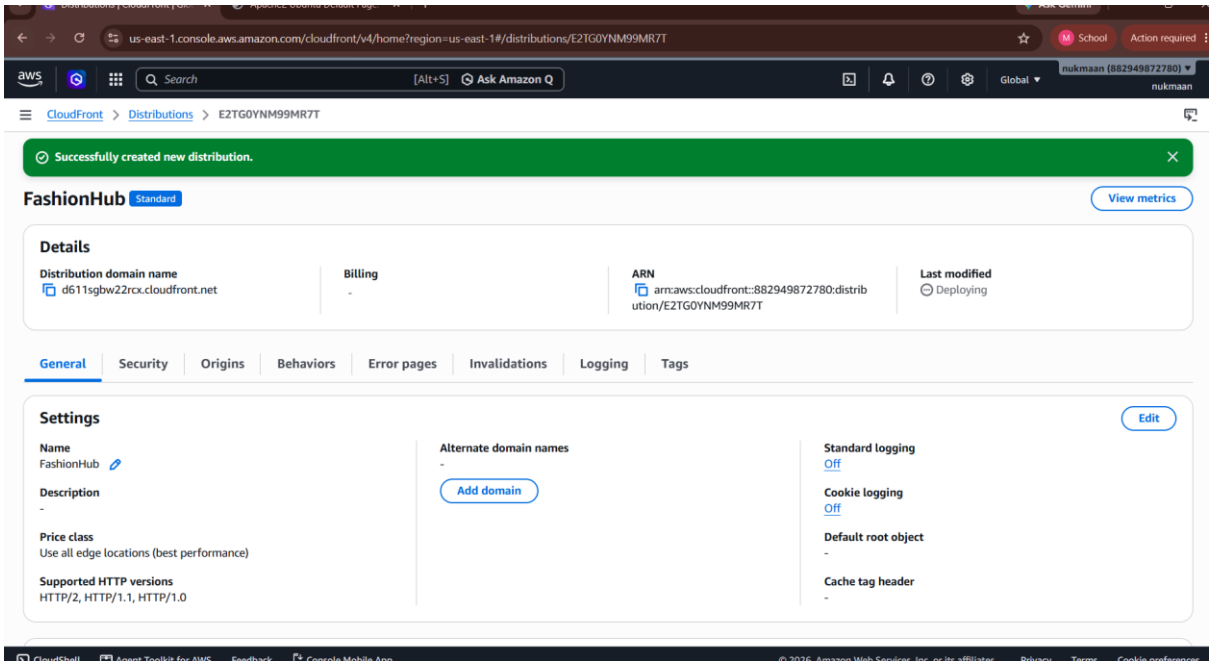
Step 9– Web Application Firewall Configuration

Configuration:
AWS WAF was not enabled because the project uses a low-cost AWS configuration and does not require advanced web application firewall protection for the demonstration.

Selected Option:
Do not enable security protections

Reason:
Disabling WAF avoids additional AWS WAF charges while maintaining the required S3, CloudFront, IAM, and EC2 functionality.

SCREENSHOT:



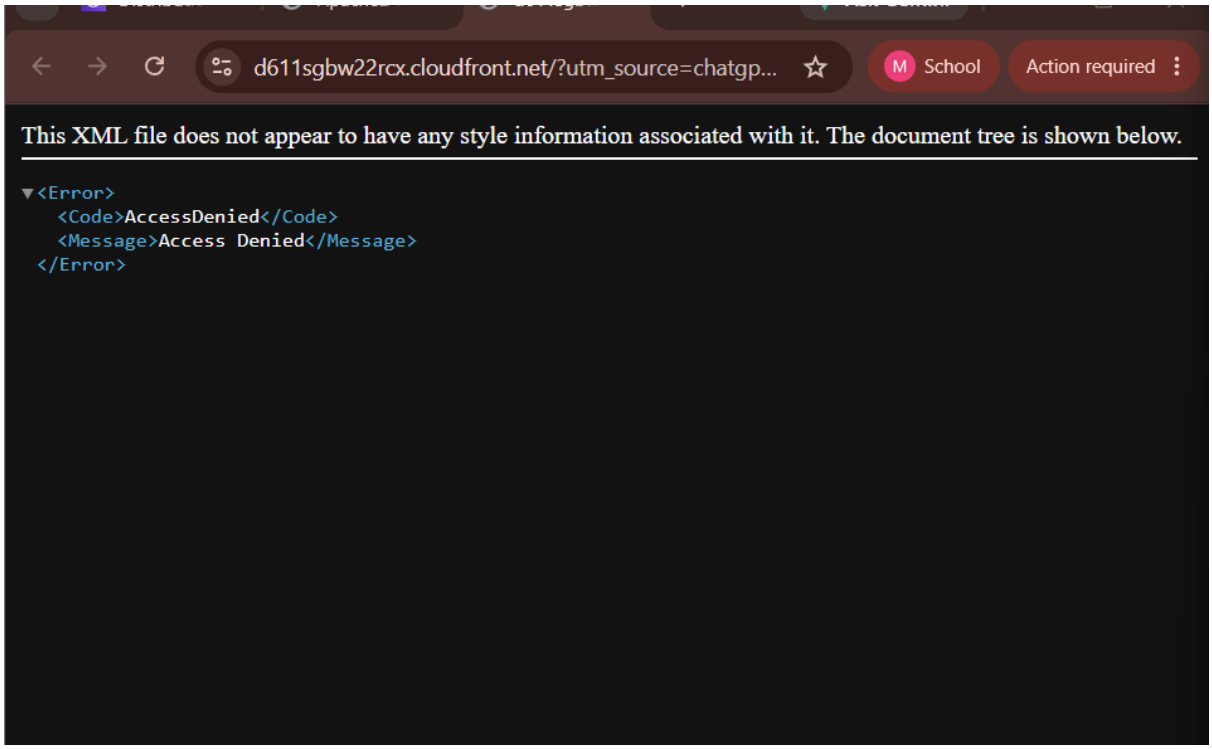
Step 10– CloudFront Distribution Testing

CloudFront Domain Name:

d611sgbw22rcx.cloudfront.net

Procedure:
The CloudFront distribution domain name was opened using HTTPS to verify the secure content delivery configuration.

SCREENSHOT:



Step 11 – Configure S3 Bucket Policy for CloudFront

The S3 bucket policy was configured to allow the CloudFront distribution to securely retrieve product images using Origin Access Control (OAC), while public access to the S3 bucket remained blocked.

SCREENSHOT:



Step 12– Display Product Images Using CloudFront

Procedure

1. Logged in to the EC2 instance using SSH.
2. Created/updated the Apache web page at:
   /var/www/html/index.html
3. Added the FashionHub webpage HTML code.
4. Added the three product images using their CloudFront URLs.
5. Restarted the Apache web server using:
   systemctl restart apache2
6. Opened the EC2 instance Public IPv4 address in a web browser.
7. Refreshed the webpage and verified that all three FashionHub images were displayed successfully.

CloudFront Image URLs

- **FashionHub Design 4**
- **FashionHub Design 3**
- **FashionHub Design 2**

Result

The FashionHub webpage was successfully hosted on the EC2 Apache web server. All three product images stored in the private S3 bucket were successfully displayed through the CloudFront distribution.

SCREENSHOT :

